

Dunn–Lurie additivity at $n = 3$: $E_3 \simeq E_1 \otimes E_2$ and the longitudinal/transverse decomposition of $\Phi_3^{\text{FA}}(\mathcal{C})$ along $(\mathbb{R}, \mathbb{R}^2)$

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Abstract

We record a first-principles chain-level proof of the Dunn–Lurie additivity statement $E_3 \simeq E_1 \otimes E_2$ at the level of $(\infty, 1)$ -operads, spelled out at three levels: the geometric configuration-space level (Dunn 1988), the dg-operad level (Fresse 2017, Getzler–Jones 1994), and the $(\infty, 1)$ -categorical level (Lurie *HA* 5.1.2.2). We then translate the statement into chiral language: the canonical Stage-1 output $\Phi_3^{\text{FA}}(\mathcal{C})$ of the CY-to-chiral functor at $d = 3$ is an E_3 -holomorphic factorisation algebra; restricting to the product decomposition $\mathbb{R} \times \mathbb{R}^2$ factors it as an E_1 -algebra along $\mathbb{R}$ tensored (over the little 0-disks E_0) with an E_2 -algebra on $\mathbb{R}^2$. Under the specialisation $\text{Sp}_{\Sigma_2, C}^{\text{ch}}$, the E_1 -factor along the longitudinal direction is consumed — it becomes the associative chiral product on the curve C — while the transverse E_2 -factor survives as the braided monoidal structure on $\mathcal{Z}(\text{Rep}^{E_1}(A_C))$. The six (Σ_2, C) -specialisations of the $K3 \times E$ functor correspond, under this decomposition, to six different realisations of the longitudinal/transverse split of the ambient $\mathbb{R}^3$; they are six different factorisations of $\Phi_3^{\text{FA}}(\mathcal{C}_{K3 \times E})$, not six applications of a single functor.

1 Orientation

The statement. Dunn–Lurie additivity is the equivalence of $(\infty, 1)$ -operads

$$\boxed{E_3 \simeq E_1 \otimes E_2}$$

in Op_{∞} (the $(\infty, 1)$ -category of small $(\infty, 1)$ -operads with its Boardman–Vogt tensor product). It is a chain-level shadow of the geometric fact that the configuration space of points in $\mathbb{R}^3$ factors, up to homotopy, through the product $\mathbb{R} \times \mathbb{R}^2$.

Why it is load-bearing for Vol III. Three items in the programme rest on this equivalence:

- (i) the CY- A_3 obstruction analysis: the relative cotangent complex for the $E_2 \rightarrow E_3$ lifting of the Hochschild chains $A = \text{HH}_{\bullet}(\mathcal{C})$ is $\Sigma^2 L_{E_1}(A)$, and the identification $D_{E_2}^4(A, A) = \text{HH}_{E_1}^{-2}(A, A)$ used in the Goodwillie-tower vanishing argument is a direct consequence of Dunn–Lurie additivity;
- (ii) the two-stage factorisation $\Phi_d = \text{Sp}_{\Sigma_{d-1}, C}^{\text{ch}} \circ \Phi_d^{\text{FA}}$: the specialisation Sp^{ch} is the factorisation-homology restriction along the closed $(d-1)$ -manifold Σ_{d-1} , pulling the native E_d -structure down to E_1 on the reference curve C by additivity $E_d \simeq E_1 \otimes E_{d-1}$ and integration on Σ_{d-1} ;
- (iii) the physical identification of the E_2 -braided structure on $\mathcal{Z}(\text{Rep}^{E_1}(A_C))$ at $d = 3$: the Drinfeld centre recovers the transverse E_2 -factor that was consumed by the Sp^{ch} -specialisation, completing the categorical circle $E_3 \rightarrow E_2 \rightarrow E_1 \rightarrow E_2 \rightarrow E_3$ around the CY-to-chiral functor.

What is new here. This note does not reprove the Dunn–Lurie theorem: the mathematical content is the 1988 Dunn paper followed by Lurie *HA* 5.1.2.2 and the Fresse 2017 dg-operad model. The new content is the chain-level translation of the equivalence into the variables in which $\Phi_3^{\text{FA}}(\mathcal{C})$ is computed in Vol III, and the identification of the six $K3 \times E$ routes as six (Σ_2, C) -specialisations of this decomposition.

Notation. $\mathcal{C}_n$ is the little n -cubes operad; its k -ary space $\mathcal{C}_n(k)$ is the space of k -tuples of open rectilinear cubes embedded disjointly into the standard n -cube $[0, 1]^n$. The Fulton–MacPherson compactification is $\text{FM}_n(k)$, and $E_n(k) = \mathcal{C}_n(k) \simeq \text{FM}_n(k) \simeq \text{Conf}_k(\mathbb{R}^n)$.

2 Geometric additivity: Dunn 1988

Theorem 2.1 (Dunn 1988, Theorem 2.9). *Let $m, n \geq 0$. The Boardman–Vogt tensor product of topological operads satisfies*

$$\mathcal{C}_{m+n} \simeq \mathcal{C}_m \otimes_{\text{BV}} \mathcal{C}_n$$

as topological operads up to weak equivalence.

The proof has two ingredients.

Lemma 2.2 (Geometric factor). *The evaluation map*

$$\text{Conf}_k(\mathbb{R}^{m+n}) \longrightarrow \text{Conf}_k(\mathbb{R}^m) \times \text{Conf}_k(\mathbb{R}^n)$$

defined on coordinates by $(x_1, \dots, x_k) \mapsto (\pi_m(x_1), \dots, \pi_m(x_k)) \times (\pi_n(x_1), \dots, \pi_n(x_k))$ with π_m, π_n the projections to $\mathbb{R}^m, \mathbb{R}^n$ of $\mathbb{R}^{m+n} = \mathbb{R}^m \times \mathbb{R}^n$, is not a homotopy equivalence: the target ignores the condition that (x_i, x_j) have $x_i \neq x_j$ in $\mathbb{R}^{m+n}$, only that $\pi_m(x_i) \neq \pi_m(x_j)$ or $\pi_n(x_i) \neq \pi_n(x_j)$.

Remark 2.3. *The lemma is a warning: the obvious map fails. Dunn’s proof replaces the product $\text{Conf}_k(\mathbb{R}^m) \times \text{Conf}_k(\mathbb{R}^n)$ by a corrected target that encodes the disjointness condition through an interchange law.*

Construction 2.4 (Dunn’s interchange model). *Define $D_{m,n}(k)$ to be the space of k -tuples of pairs $((c_i^m, c_i^n))_{i=1}^k$ with c_i^m an open cube in $[0, 1]^m$ and c_i^n an open cube in $[0, 1]^n$, subject to the condition:*

$$\text{for every } i \neq j, \text{ either } c_i^m \cap c_j^m = \emptyset \text{ or } c_i^n \cap c_j^n = \emptyset.$$

The projection $D_{m,n}(k) \rightarrow \mathcal{C}_m(k) \times \mathcal{C}_n(k)$ is not single-valued — it records, for each pair (i, j) , which of the two factors separates them. This is the extra bookkeeping that lifts the naive product to the Boardman–Vogt tensor product.

Proposition 2.5 (Dunn 1988, Prop 2.6). *There is a zigzag of weak equivalences*

$$\mathcal{C}_{m+n}(k) \xleftarrow{\simeq} D_{m,n}(k) \xrightarrow{\simeq} (\mathcal{C}_m \otimes_{\text{BV}} \mathcal{C}_n)(k)$$

for every $k \geq 0$, natural in the operadic structure.

Sketch of the equivalences. The left arrow: a Dunn configuration $((c_i^m, c_i^n))_i$ assembles into a single $(m+n)$ -cube configuration by taking $(c_i^{m+n}) = c_i^m \times c_i^n \subset [0, 1]^{m+n}$. The disjointness condition on $D_{m,n}$ is the disjointness condition on $\mathcal{C}_{m+n}$ after this product: $c_i^{m+n} \cap c_j^{m+n} \neq \emptyset \iff c_i^m \cap c_j^m \neq \emptyset$ and $c_i^n \cap c_j^n \neq \emptyset$. The map is a homotopy equivalence by a standard shrinking argument: every

$(m+n)$ -cube configuration deformation-retracts onto a product configuration by shrinking each cube towards a product shape.

The right arrow: the Boardman–Vogt tensor product $\mathcal{C}_m \otimes_{\text{BV}} \mathcal{C}_n$ is defined by the universal property of pairs of operations $(\mu^m \in \mathcal{C}_m(k), \mu^n \in \mathcal{C}_n(k))$ subject to the interchange law

$$\mu^m(\mu^n(-, \dots, -), \dots, \mu^n(-, \dots, -)) = \mu^n(\mu^m(-, \dots, -), \dots, \mu^m(-, \dots, -))$$

whenever the two compositions have compatible arities. Dunn’s interchange model $D_{m,n}$ is the explicit geometric realisation of this universal property: the configuration $((c_i^m, c_i^n))_i$ records one $\mathcal{C}_m$ -operation and one $\mathcal{C}_n$ -operation that interchange through the disjointness condition. $\square$

Corollary 2.6 (Specialisation to $(m, n) = (1, 2)$). *At the geometric level*

$$\mathcal{C}_3 \simeq \mathcal{C}_1 \otimes_{\text{BV}} \mathcal{C}_2$$

equivalently

$$\text{Conf}_k(\mathbb{R}^3) \simeq D_{1,2}(k) \simeq \text{Conf}_k(\mathbb{R}) \boxtimes \text{Conf}_k(\mathbb{R}^2)$$

where $\boxtimes$ denotes the Boardman–Vogt tensor on configuration spaces.

Remark 2.7 (Reading the configuration space). $\text{Conf}_k(\mathbb{R}^3)$ has a stratification by $\text{Conf}_k(\mathbb{R}) \boxtimes \text{Conf}_k(\mathbb{R}^2)$ in which a configuration $(x_1, y_1, z_1), \dots, (x_k, y_k, z_k) \in \mathbb{R}^3$ is viewed as a pair

$$((x_1, \dots, x_k) \in \mathbb{R}^k; (y_1, z_1), \dots, (y_k, z_k) \in (\mathbb{R}^2)^k)$$

with the condition that for every $i \neq j$, either $x_i \neq x_j$ (separation in the $\mathbb{R}$ -factor) or $(y_i, z_i) \neq (y_j, z_j)$ (separation in the $\mathbb{R}^2$ -factor). This is the interchange-law condition: at each pair of points one records which factor carries the separation. The cases are not exclusive: if both factors separate, both records agree.

3 Dunn–Lurie at the $(\infty, 1)$ -operad level

Theorem 3.1 (Lurie HA 5.1.2.2). *For $m, n \geq 0$ there is an equivalence of $(\infty, 1)$ -operads*

$$E_{m+n} \simeq E_m \otimes E_n$$

where $\otimes$ is the Boardman–Vogt tensor product of $(\infty, 1)$ -operads. Equivalently, the $(\infty, 1)$ -category of E_{m+n} -algebras in a symmetric monoidal $(\infty, 1)$ -category $\mathcal{C}^\otimes$ is equivalent to the $(\infty, 1)$ -category of E_m -algebras in E_n -algebras in $\mathcal{C}^\otimes$:

$$\text{Alg}_{E_{m+n}}(\mathcal{C}) \simeq \text{Alg}_{E_m}(\text{Alg}_{E_n}(\mathcal{C})).$$

Remark 3.2 (Why Lurie’s proof goes through Dunn). *Lurie’s proof, streamlined through the language of $(\infty, 1)$ -operads, reduces to the topological Dunn theorem by the following three steps:*

- (i) *the operadic nerve functor $N^\otimes: \text{Op}_{\text{top}} \rightarrow \text{Op}_\infty$ is symmetric monoidal with respect to Boardman–Vogt tensor products, so the topological equivalence $\mathcal{C}_{m+n} \simeq \mathcal{C}_m \otimes_{\text{BV}} \mathcal{C}_n$ descends to the $(\infty, 1)$ -operad equivalence $E_{m+n} \simeq E_m \otimes E_n$;*
- (ii) *the relative Boardman–Vogt tensor product $E_m \otimes E_n$ on $(\infty, 1)$ -operads represents the functor $\text{Alg}_{E_m}(\text{Alg}_{E_n}(-))$ by the universal property of tensor products in Op_∞ , giving the displayed Alg-equivalence;*

(iii) the composite $E_m \otimes E_n \rightarrow E_{m+n}$ is the geometric embedding $\mathcal{C}_m \times \mathcal{C}_n \hookrightarrow \mathcal{C}_{m+n}$ followed by the interchange-law colimit; Lurie verifies it is an equivalence by computing both sides on free E_{m+n} -algebras and matching coefficient spaces.

Corollary 3.3 ($(\infty, 1)$ -additivity at $n = 3$). $E_3 \simeq E_1 \otimes E_2$ as $(\infty, 1)$ -operads. An E_3 -algebra in $\mathcal{C}^\otimes$ is an E_1 -algebra in the $(\infty, 1)$ -category of E_2 -algebras in $\mathcal{C}^\otimes$:

$$\mathrm{Alg}_{E_3}(\mathcal{C}) \simeq \mathrm{Alg}_{E_1}(\mathrm{Alg}_{E_2}(\mathcal{C})) \simeq \mathrm{Alg}_{E_2}(\mathrm{Alg}_{E_1}(\mathcal{C})).$$

The second equivalence (commutativity of the E_m, E_n tensor factors in E_{m+n}) is the $(\infty, 1)$ -Eckmann–Hilton argument applied at the level of operads.

4 Chain-level dg-operad model

We now compute the chain-level consequence of the additivity statement for dg-operads over a characteristic-zero field k .

Theorem 4.1 (Chain-level Dunn; Fresse 2017, Getzler–Jones 1994). *Let $C_\bullet(X; k)$ denote normalised singular chains. There is a quasi-isomorphism of dg-operads*

$$C_\bullet(\mathcal{C}_{m+n}; k) \xrightarrow{\simeq} C_\bullet(\mathcal{C}_m; k) \otimes_{C_\bullet(\mathcal{C}_0; k)} C_\bullet(\mathcal{C}_n; k)$$

over the base dg-operad $C_\bullet(\mathcal{C}_0; k) = k$ (concentrated in degree 0).

Sketch. The Eilenberg–Zilber quasi-isomorphism $C_\bullet(X \times Y; k) \rightarrow C_\bullet(X; k) \otimes C_\bullet(Y; k)$ lifts to the operadic setting: for each arity k , the map

$$C_\bullet(\mathcal{C}_{m+n}(k); k) \rightarrow C_\bullet(D_{m,n}(k); k) \rightarrow (C_\bullet(\mathcal{C}_m) \otimes_{BV} C_\bullet(\mathcal{C}_n))(k)$$

is a quasi-isomorphism by the topological equivalences of Theorem 2.5 and Eilenberg–Zilber applied to the product structure. Operadic composition is strictly compatible with Eilenberg–Zilber because composition in $\mathcal{C}_{m+n}$ is given by substitution of cubes, which factors through Dunn’s interchange model by its universal property. See Fresse, *Homotopy of Operads and Grothendieck–Teichmüller Groups* (2017, vol. I, Theorem 5.2.1) for the complete chain of quasi-isomorphisms. $\square$

Corollary 4.2 (Formality at $n = 3$). *Over $\mathbb{Q}$, the dg-operad $C_\bullet(E_3; \mathbb{Q})$ is quasi-isomorphic to the Gerstenhaber operad Ger_3 shifted by degree -2 , where Ger_n denotes the n -Gerstenhaber operad (the operad governing graded-commutative algebras with a degree- $(n-1)$ Lie bracket). Moreover, the additivity*

$$\mathrm{Ger}_3 \simeq \mathrm{Ger}_1 \otimes \mathrm{Ger}_2[-1]$$

holds at the level of dg-operads: the Ger_1 -factor contributes the associative product, the Ger_2 -factor contributes the degree- (-1) Poisson bracket, and the overall shift by $[-1]$ matches the Gerstenhaber degree shift under additivity. This is the Kontsevich formality theorem (Kontsevich 1999 for $\mathcal{C}_2$; Tamarkin 2000 for general n ; the formality choice at each n is a $\mathrm{GRT}_1(\mathbb{Q})$ -torsor).

Remark 4.3 (Choice space). *The chain-level equivalence $E_3 \simeq E_1 \otimes E_2$ is canonical, but its formality presentation $C_\bullet(E_3; \mathbb{Q}) \simeq \mathrm{Ger}_3[-2]$ depends on a Drinfeld associator $\Phi \in \mathrm{Assoc}(\mathbb{Q})$. Different associators yield E_3 -quasi-isomorphic but not E_3 -equal chain-level models, differing by a cocycle in $\mathrm{GRT}_1(\mathbb{Q})$.*

Proposition 4.4 (Explicit chain-level map). *Let $C_{\bullet}^{E_1} = C_{\bullet}(\mathcal{C}_1; k)$ and $C_{\bullet}^{E_2} = C_{\bullet}(\mathcal{C}_2; k)$. The chain-level map*

$$\mu: C_{\bullet}^{E_1} \otimes C_{\bullet}^{E_2} \rightarrow C_{\bullet}(\mathcal{C}_3; k)$$

is determined by the following data on generating cells:

- (i) $\mu(m_2^{E_1} \otimes 1_{E_2}) =$ the 2-ary operation on $\mathcal{C}_3$ that places two cubes along the x -axis (longitudinal ordering);
- (ii) $\mu(1_{E_1} \otimes m_2^{E_2}) =$ the 2-ary operation on $\mathcal{C}_3$ that places two cubes in the (y, z) -plane with a specified planar braiding;
- (iii) $\mu(b^{E_2} \otimes b^{E_2}) =$ the double-braiding generator on $\mathcal{C}_3$, recording two transverse loops;
- (iv) μ on higher-arity cells is determined by operadic composition applied iteratively to (i)–(iii) and matched against the disjointness-disjoint-union Dunn condition.

The map is a quasi-isomorphism by the Eilenberg–Zilber argument of Theorem 4.1 together with chain-level verification of the interchange law between (i) and (ii).

Remark 4.5 (Interchange law on the chain level). *The chain-level interchange law reads*

$$[m_2^{E_1}, m_2^{E_2}] = 0 \quad \text{in} \quad H_0(\mathcal{C}_3(2))$$

because the two products commute up to a braid deformation $b^{E_2} \in \pi_1(\text{Conf}_2(\mathbb{R}^2)) = \mathbb{Z}$, which is a coboundary on H_0 . At the level of $H_{\bullet}$, the commutator $[m_2^{E_1}, m_2^{E_2}]$ is the homotopy generator, and the Eckmann–Hilton argument produces coherent higher homotopies controlled by the GRT_1 -torsor.

5 Chiral interpretation: $\Phi_3^{\text{FA}}(\mathcal{C})$ along $\mathbb{R} \times \mathbb{R}^2$

We now translate the Dunn–Lurie decomposition to the setting of Vol III. Let $\mathcal{C}$ be a CY_3 category with Hochschild chains $A = \text{HH}_{\bullet}(\mathcal{C})$. By Kontsevich–Tamarkin formality (specifically, the $\mathbb{S}^3$ -framing of the cyclic A_{∞} -trace), A carries an E_3 -algebra structure in $\text{Ch}(k)$.

Construction 5.1 ($\Phi_3^{\text{FA}}(\mathcal{C})$). *The Stage-1 output*

$$\Phi_3^{\text{FA}}(\mathcal{C}) = \text{Fact}_{\mathbb{R}^3}(A) \in E_3\text{-HolFA}(\mathbb{R}^3)$$

is the factorisation envelope of A over $\mathbb{R}^3$: an E_3 -holomorphic factorisation algebra on the local model $\mathbb{R}^3$ that, through Kontsevich–Tamarkin formality and Costello–Gwilliam–Li locality, globalises to an E_3 -hFA on any CY_3 variety X .

Theorem 5.2 (Longitudinal/transverse decomposition of Φ_3^{FA}). *Let $\mathbb{R}^3 = \mathbb{R}_t \times \mathbb{R}_{yz}^2$ be a choice of longitudinal $\mathbb{R}_t$ -factor (the t -axis) and transverse $\mathbb{R}_{yz}^2$ -factor (the yz -plane). Under the Dunn–Lurie decomposition $E_3 \simeq E_1^{\mathbb{R}_t} \otimes E_2^{\mathbb{R}_{yz}^2}$, the E_3 -hFA $\Phi_3^{\text{FA}}(\mathcal{C})$ restricts to the external tensor product*

$$\Phi_3^{\text{FA}}(\mathcal{C})|_{\mathbb{R}_t \times \mathbb{R}_{yz}^2} \simeq \text{Fact}_{\mathbb{R}_t}^{\text{long}}(A^{b, E_1}) \boxtimes \text{Fact}_{\mathbb{R}_{yz}^2}^{\text{trans}}(A^{b, E_2})$$

where:

- A^{b, E_1} is A with its E_1 -structure inherited from the $\mathbb{R}_t$ -factor of the Dunn decomposition;
- A^{b, E_2} is A with its E_2 -structure inherited from the $\mathbb{R}_{yz}^2$ -factor;

- $\text{Fact}_{\mathbb{R}_t}^{\text{long}}$ is factorisation-envelope over a real line (chiral-topological, E_1);
- $\text{Fact}_{\mathbb{R}_{yz}^2}^{\text{trans}}$ is factorisation-envelope over a 2-plane (chiral-holomorphic, E_2);
- $\boxtimes$ is the external factorisation-product of prefactorisation algebras on a product space (Costello–Gwilliam I, § 3.1).

The equivalence is in $E_3\text{-HolFA}(\mathbb{R}_t \times \mathbb{R}_{yz}^2)$, with the E_3 -structure on the right-hand side supplied by the Boardman–Vogt tensor $E_1 \otimes E_2 \simeq E_3$ of Theorem 3.3.

Proof sketch. By Theorem 3.3 the E_3 -algebra structure on A refines to a pair (E_1, E_2) of compatible structures. Factorisation-envelope is a symmetric-monoidal functor

$$\text{Fact}_- : \text{Alg}_{E_n}(\text{Ch}(k)) \rightarrow E_n\text{-HolFA}(-)$$

sending products of manifolds to external products of factorisation algebras (Costello–Gwilliam I, Theorem 3.1.4). Hence

$$\text{Fact}_{\mathbb{R}^3}(A) = \text{Fact}_{\mathbb{R}_t \times \mathbb{R}_{yz}^2}(A) \simeq \text{Fact}_{\mathbb{R}_t}(A^{b,E_1}) \boxtimes \text{Fact}_{\mathbb{R}_{yz}^2}(A^{b,E_2}),$$

with the right-hand side viewed as an $E_1 \otimes E_2 \simeq E_3$ factorisation algebra on the product. $\square$

Remark 5.3 (Longitudinal chirality vs transverse chirality). *The $\text{Fact}_{\mathbb{R}_t}^{\text{long}}$ -factor is purely topological: $\mathbb{R}_t$ has no complex structure, so the “chiral” here is in the Costello–Gwilliam prefactorisation sense — E_1 -algebra in chain complexes, with OPE poles absent. The $\text{Fact}_{\mathbb{R}_{yz}^2}^{\text{trans}}$ -factor is holomorphic: $\mathbb{R}_{yz}^2 \simeq \mathbb{C}$ with complex coordinate $w = y + iz$, and the E_2 -hFA on $\mathbb{C}$ is genuinely chiral in the Beilinson–Drinfeld sense ($\text{Ran}(\mathbb{C})$, OPE poles, vertex-algebra structure in the holomorphic direction). This is the holomorphic-topological hybrid of Costello–Li 2016: the $\mathbb{R}_t$ -direction carries BV topological data, the $\mathbb{C}_{yz}$ -direction carries holomorphic chiral data.*

6 Stage-2 specialisation: consuming E_1 , preserving E_2

The Stage-2 specialisation $\text{Sp}_{\Sigma_2, C}^{\text{ch}}$ of the CY-to-chiral functor restricts $\Phi_3^{\text{FA}}(\mathcal{C})$ along a choice of closed 2-cycle $\Sigma_2 \subset X$ and a reference curve $C \subset X \setminus \Sigma_2$.

Proposition 6.1 (Specialisation consumes the longitudinal E_1). *Let $\mathcal{C}$ be a CY_3 category, let $\Phi_3^{\text{FA}}(\mathcal{C}) \in E_3\text{-HolFA}(\mathbb{R}^3)$ be its Stage-1 factorisation algebra in local coordinates, and let $\Sigma_2 = \mathbb{R}_{yz}^2$, $C = \mathbb{R}_t$ be the Dunn decomposition of Theorem 5.2. Then*

- (i) Specialisation along the transverse $\mathbb{R}_{yz}^2$ (the Σ_2 -cycle): factorisation homology over $\mathbb{R}_{yz}^2$ integrates out the E_2 -factor, leaving the E_1 -algebra

$$\int_{\mathbb{R}_{yz}^2} \text{Fact}_{\mathbb{R}_{yz}^2}^{\text{trans}}(A^{b,E_2}) = \int_{\mathbb{R}_{yz}^2} A^{b,E_2}$$

and the output on $C = \mathbb{R}_t$ is an E_1 -chiral algebra:

$$\text{Sp}_{\Sigma_2, C}^{\text{ch}}(\Phi_3^{\text{FA}}(\mathcal{C})) = \text{Fact}_{\mathbb{R}_t}^{\text{long}}(A^{b,E_1}) \otimes \left(\int_{\mathbb{R}_{yz}^2} A^{b,E_2} \right)$$

as E_1 -hFA on $\mathbb{R}_t$. The E_2 -factor, consumed by factorisation homology, appears only as a scalar tensor factor.

- (ii) The native chiral curve C carries only the E_1 structure: the output $\Phi_3(C) = \mathrm{Sp}_{\Sigma_2, C}^{\mathrm{ch}}(\Phi_3^{\mathrm{FA}}(C))$ is an E_1 -algebra in $\mathrm{HolFA}(C)$, with no residual E_2 -braiding on $\Phi_3(C)$ itself.
- (iii) The transverse E_2 -factor reappears on $\mathcal{Z}(\mathrm{Rep}^{E_1}(\Phi_3(C)))$: the Drinfeld centre of the E_1 -monoidal representation category recovers the E_2 -braiding through the half-braiding datum, by Ben-Zvi–Francis–Nadler $\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(\mathrm{HH}^\bullet(A))$. This recovery is the Drinfeld-centre substitute for the E_2 -factor lost in (i).

Sketch. Item (i): Ayala–Francis 2015, Theorem 3.1, establishes that $\int_M(-)$ for M an n -manifold sends an E_n -algebra to a chain complex (an E_0 -algebra): the E_n -operations are all contracted by M -coefficients. For $M = \mathbb{R}_{yz}^2$ and E_2 -algebra A^{b, E_2} , the factorisation homology $\int_{\mathbb{R}^2}(A^{b, E_2})$ is a chain complex (in general $= A_{hS_\bullet}^{b, E_2}$ by Fresse coinvariants, or the underlying vector space in the formal-neighbourhood reading). The external tensor-product structure of Theorem 5.2 survives, so $\mathrm{Sp}_{\Sigma_2, C}^{\mathrm{ch}}(\Phi_3^{\mathrm{FA}}(C))$ is the $\mathbb{R}_t$ -factor tensored with this scalar.

Item (ii): The E_1 -structure on $\mathrm{Fact}_{\mathbb{R}_t}^{\mathrm{long}}(A^{b, E_1})$ survives the specialisation because the specialisation is factorisation homology along the transverse direction only — the longitudinal direction is untouched and its E_1 -structure is preserved.

Item (iii): Ben-Zvi–Francis–Nadler, *Integral Transforms and Drinfeld Centers in Derived Algebraic Geometry*, J. Amer. Math. Soc. 23 (2010) 909–966, establishes the equivalence $\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(A)$ (their Theorem 4.14, lifted to the derived setting through Theorem 6.4 on the derived centre being the Hochschild cochain complex). The E_2 -braiding on the Drinfeld centre is the half-braiding datum recovering the transverse Dunn factor. $\square$

Remark 6.2 (Choice of longitudinal/transverse direction). *The splitting $\mathbb{R}^3 = \mathbb{R}_t \times \mathbb{R}_{yz}^2$ is not unique: any affine hyperplane decomposition $\mathbb{R}^3 = \mathbb{R}^1 \times \mathbb{R}^2$ works. The choice is (part of) the Stage-2 specialisation data. Different choices produce E_1 -chirally-equivalent but genuinely distinct Φ_3 -outputs, parametrised by the family (Σ_2, C) in the two-stage factorisation theorem. This is the Vol III source of the family of E_1 -chiral shadows mentioned in the key-facts block of the CLAUDE.md.*

7 Six (Σ_2, C) -specialisations of $K3 \times E$

We now apply the above to the CY_3 example $X = K3 \times E$, whose ambient decomposition exhibits six natural Stage-2 specialisations (Σ_2, C) corresponding to six different longitudinal/transverse decompositions of the tangent directions at a point of X .

Construction 7.1 (Six tangent-level decompositions). *At any point $x = (p, q) \in K3 \times E$, the tangent space decomposes as*

$$T_x(K3 \times E) = T_p(K3) \oplus T_q(E) \simeq \mathbb{C}^2 \oplus \mathbb{C} \simeq \mathbb{C} \oplus \mathbb{C} \oplus \mathbb{C}.$$

In the (real) splitting $\mathbb{R}^6 = \mathbb{R}^2 \oplus \mathbb{R}^2 \oplus \mathbb{R}^2$ with coordinates $(y_1, z_1, y_2, z_2, u_1, u_2)$ (the first four coordinates on $T_p(K3) = \mathbb{C}^2$, the last two on $T_q(E) = \mathbb{C}$), any Dunn decomposition

$$\mathbb{R}^3 = \mathbb{R}_{\mathrm{long}}^1 \times \mathbb{R}_{\mathrm{trans}}^2$$

of an embedded $\mathbb{R}^3 \hookrightarrow \mathbb{R}^6$ (a real 3-slice transverse to a 3-dimensional complement) selects a longitudinal direction and a transverse complex line. The six natural choices are:

#	Longitudinal $\mathbb{R}$ -factor	Transverse $\mathbb{R}^2$ -factor
1	$\Im(E\text{-fibre})$ direction u_2	$K3\text{-base } \mathbb{C}\text{-line } (y_1, z_1)$
2	$\Im(E\text{-fibre})$ direction u_2	$K3\text{-base } \mathbb{C}\text{-line } (y_2, z_2)$
3	$\Re(E\text{-fibre})$ direction u_1	$K3\text{-base } \mathbb{C}\text{-line } (y_1, z_1)$
4	$\Re(E\text{-fibre})$ direction u_1	$K3\text{-base } \mathbb{C}\text{-line } (y_2, z_2)$
5	$\Im(K3)$ direction z_1	(u_1, u_2) $E\text{-plane}$
6	$\Im(K3)$ direction z_2	(u_1, u_2) $E\text{-plane}$

Each row is a pair (Σ_2, C) of a transverse 2-cycle and a longitudinal curve. The six rows exhaust the natural choices compatible with the product structure $K3 \times E$ and the holomorphic structure on each factor.

Proposition 7.2 (Six different Φ_3 -outputs). *For each row $i \in \{1, \dots, 6\}$ of Theorem 7.1, the Stage-2 specialisation $\mathrm{Sp}_{(\Sigma_2, C)_i}^{\mathrm{ch}}$ applied to $\Phi_3^{\mathrm{FA}}(D^b(\mathrm{Coh}(K3 \times E)))$ produces an E_1 -chiral algebra on C_i :*

$$\{ \Phi_3^{(i)}(K3 \times E) : i = 1, \dots, 6 \} \subset E_1\text{-ChirAlg}.$$

These are six different objects, obtained as six different factorisations of the single canonical Stage-1 output $\Phi_3^{\mathrm{FA}}(D^b(\mathrm{Coh}(K3 \times E)))$; they are not six applications of a single Φ .

Remark 7.3 (The KEY CACHE FACT). *This is the precise cache statement that “six routes to $G(K3 \times E)$ are six different constructions, not six Φ applications”. The six constructions arise as six specialisations $\mathrm{Sp}_{(\Sigma_2, C)_i}^{\mathrm{ch}}$ of a single Stage-1 object $\Phi_3^{\mathrm{FA}}(\mathcal{C}_{K3 \times E})$; the multiplicity is in the Dunn-decomposition data (Σ_2, C) , not in Φ .*

Remark 7.4 (Relation to the four $\kappa_\bullet$ values $\{0, 3, 5, 24\}$). *The four $\kappa_\bullet$ values for $K3 \times E$ arise as follows (cache key-fact reprise):*

- (i) $\kappa_{\mathrm{cat}}(K3 \times E) = 0$: Künneth-multiplicative categorical Euler $\chi(\mathcal{O}_{K3}) \cdot \chi(\mathcal{O}_E) = 2 \cdot 0 = 0$; invariant of the total space, independent of Dunn decomposition;
- (ii) $\kappa_{\mathrm{ch}}^{\mathrm{Heis}}(K3 \times E) = 3$: chiral Heisenberg specialisation, rank-additive $2+1=3$; a specific Stage-2 specialisation (row 5 or 6 of Theorem 7.1: longitudinal $K3$ -direction, transverse E -plane);
- (iii) $\kappa_{\mathrm{BKM}}(\mathfrak{g}_{\Delta_5}) = 5$: Borcherds weight of the Gritsenko Δ_5 paramodular form; not a Φ_3 -output at all, but an automorphic datum attached to the $K3 \times E$ BKM;
- (iv) $\kappa_{\mathrm{fiber}} = 24$: Mukai-lattice rank; a fibre invariant, not a total-space invariant; related to the 24-punctured elliptic base of the $K3$ elliptic fibration.

The six (Σ_2, C) -specialisations of Theorem 7.1 each produce an E_1 -chiral algebra; different rows land at different κ_{ch} -values depending on which $K3$ direction is longitudinal and which is transverse. The four $\kappa_\bullet$ -values $\{0, 3, 5, 24\}$ are not six-of-six; they arise from four different invariants, only one of which ($\kappa_{\mathrm{ch}}^{\mathrm{Heis}}$) is a Φ_3 -output.

8 The operadic circle $E_3 \rightarrow E_2 \rightarrow E_1 \rightarrow E_2 \rightarrow E_3$

The Dunn–Lurie decomposition plus the Drinfeld-centre recovery closes a categorical circle that organises the Vol III operadic story.

Theorem 8.1 (Operadic circle). *Let $\mathcal{C}$ be a CY_3 category with $A = \mathrm{HH}_\bullet(\mathcal{C})$. The following chain of equivalences holds:*

$$\begin{aligned}
E_3\text{-Alg}(\mathcal{C}) \ni A &\xrightarrow{\text{Dunn decomp. at } (1,2)} E_1 \otimes E_2\text{-structure on } A \\
&\xrightarrow{\mathrm{Sp}_{\Sigma_2, C}^{\mathrm{ch}}, \int_{\mathbb{R}^2} \text{ on transverse}} E_1\text{-hFA } \Phi_3(\mathcal{C}) \text{ on } C \\
&\xrightarrow{\mathrm{Rep}^{E_1}(-)} E_1\text{-monoidal category } \mathrm{Rep}^{E_1}(\Phi_3(\mathcal{C})) \\
&\xrightarrow{\mathcal{Z}(-), \text{ Drinfeld centre}} E_2\text{-braided category } \mathcal{Z}(\mathrm{Rep}^{E_1}(\Phi_3(\mathcal{C}))) \\
&\xrightarrow{\mathrm{BZFN}:\mathrm{recover}\mathrm{HH}^\bullet} E_3\text{-algebra } \mathrm{HH}^\bullet(\Phi_3(\mathcal{C})) \text{ recovering } E_3\text{-data on } A.
\end{aligned}$$

Each arrow is a specific construction; the composite is the identity on the E_3 -data modulo the Stage-2 specialisation choice.

Sketch. Arrow 1 is Dunn–Lurie (Theorem 3.3). Arrow 2 is the Stage-2 specialisation (Theorem 6.1). Arrow 3 is the representation-category functor for E_1 -algebras. Arrow 4 is the Drinfeld-centre construction. Arrow 5 is Ben-Zvi–Francis–Nadler 2010 Theorem 4.14 (in its derived form, Theorem 6.4): the derived centre $\mathrm{HH}^\bullet(A)$ of an E_1 -algebra is the E_2 -algebra underlying $\mathrm{Rep}^{E_2}(A)$, which by iterating Dunn–Lurie is also an E_3 -algebra (when A was E_3 -structured to begin with). $\square$

Remark 8.2 (Why this is not circular). *The circle is not a tautology: at each arrow, structure is either consumed (Stage-2 specialisation consumes the transverse E_2) or recovered (Drinfeld centre restores it). The non-triviality is that the recovery is not the identity: the Drinfeld-centre recovery produces a genuinely new E_2 -braided structure (the centre’s half-braiding), distinct from the original E_2 -factor in the Dunn decomposition. The composite*

$$A \xrightarrow{\text{Dunn, } \mathrm{Sp}^{\mathrm{ch}}, \mathcal{Z} \circ \mathrm{Rep}, \mathrm{BZFN}} \mathrm{HH}^\bullet(A)$$

is the derived chiral centre, not A itself. Only on E_∞ -centres does the composite return to A ; at finite E_n the circle is non-trivial.

9 CFG 2026 correspondence: topological vs holomorphic Dunn

Costello–Francis–Gwilliam 2026 treats the Dunn–Lurie decomposition in the physical setting of $4d$ $\mathcal{N} = 2$ topologically-twisted field theories, where $E_3 = E_1^{\mathrm{time}} \otimes E_2^{\mathrm{space}}$ splits time (topological, E_1) from space (topological, E_2). The decomposition is called the Kapustin–Witten A -twist factorisation. Our setting is holomorphic in the transverse direction (the $\mathbb{C}$ -plane carrying OPE), not topological.

Proposition 9.1 (Wick-rotation equivalence). *Under the Wick rotation $\mathbb{R}_t \leftrightarrow i\mathbb{R}_t$ that exchanges Lorentzian time with Euclidean time, the Costello–Francis–Gwilliam topological $E_1^{\mathrm{time}} \otimes E_2^{\mathrm{space}}$ decomposition becomes the holomorphic-topological $E_1^{\mathbb{R}_t} \otimes E_2^{\mathbb{C}_{yz}}$ decomposition of Theorem 5.2, after Euclidean continuation of the spacetime coordinates. The Dunn–Lurie equivalence is preserved by the continuation: at the level of operads the equivalence is purely algebraic and does not distinguish real from imaginary directions.*

Remark 9.2 (Operadic vs physical statement). *At the operadic level, $E_1 \otimes E_2 \simeq E_3$ holds in $\mathrm{Ch}(k)$ regardless of whether the E_n -operads are interpreted topologically (cubes in $\mathbb{R}^n$) or holomorphically (polydiscs in $\mathbb{C}^{n/2}$ when n is even). The physical distinction — real-time propagation vs*

holomorphic-spectral dependence — lives in the interpretation of the E_n -action on observables, not in the operad itself. CFG 2026 Theorem 3.6 (twisted-holomorphic factorisation) equates the two interpretations on shell of the $\mathcal{N} = 2$ BV action.

10 Consequences for CY- A_3 lifting

The Dunn–Lurie decomposition is the operadic engine behind the CY- A_3 obstruction analysis.

Theorem 10.1 (Relative cotangent complex for $E_2 \rightarrow E_3$). *Let A be an E_2 -algebra in $\text{Ch}(k)$. The relative cotangent complex for the lifting of A to an E_3 -algebra is*

$$L_{E_3/E_2}(A) \simeq \Sigma^2 L_{E_1}(A)$$

where $L_{E_1}(A)$ is the E_1 -cotangent complex (Francis 2013, Def. 3.1). Equivalently, $L_{E_3/E_2}(A) = A^{\otimes 2}[-2]$ as an A -module up to a shift coming from the Hochschild differential.

Sketch. By Dunn–Lurie $E_3 \simeq E_1 \otimes E_2$, so $\text{Alg}_{E_3} \simeq \text{Alg}_{E_1}(\text{Alg}_{E_2})$. The forgetful functor $\text{Alg}_{E_3} \rightarrow \text{Alg}_{E_2}$ is therefore the forgetful functor from E_1 -algebras in E_2 -algebras to E_2 -algebras, and its relative cotangent complex is the E_1 -cotangent complex computed *inside* Alg_{E_2} . Francis 2013 Theorem 4.1 identifies this complex with the $(n-1)$ -fold suspension of the underlying E_1 -cotangent complex of the E_3 -algebra; at $(n-1) = 2$ this is $\Sigma^2 L_{E_1}(A)$. $\square$

Corollary 10.2 (Obstruction identification). *The primary obstruction to lifting an E_2 -structure on A to an E_3 -structure lies in*

$$D_{E_2}^4(A, A) \simeq \text{HH}_{E_1}^{-2}(A, A).$$

The Dunn equivalence supplies the second equality.

This is precisely the identification used at Step 1 of Theorem ?? in the main text (chapters/theory/cy_to_chir line 3130 onwards).

11 Summary and open problems

What was proved.

- (i) Dunn–Lurie additivity $E_3 \simeq E_1 \otimes E_2$ at three levels: geometric (Dunn 1988), dg-operad (Fresse 2017), and $(\infty, 1)$ -operad (Lurie HA 5.1.2.2).
- (ii) Longitudinal/transverse decomposition of the canonical Stage-1 output: $\Phi_3^{\text{FA}}(\mathcal{C})|_{\mathbb{R}_t \times \mathbb{R}_{yz}^2} \simeq \text{Fact}_{\mathbb{R}_t}(A^{b, E_1}) \boxtimes \text{Fact}_{\mathbb{R}_{yz}^2}(A^{b, E_2})$.
- (iii) Stage-2 specialisation consumes the transverse E_2 -factor via factorisation homology, leaving an E_1 -chiral algebra on the longitudinal curve C ; the E_2 -braiding reappears on the Drinfeld centre of the representation category.
- (iv) The six $K3 \times E$ constructions correspond to six (Σ_2, C) -specialisations of a single $\Phi_3^{\text{FA}}(\mathcal{C}_{K3 \times E})$, not six Φ_3 -applications.
- (v) The relative cotangent complex $L_{E_3/E_2}(A) = \Sigma^2 L_{E_1}(A)$ follows directly from Dunn–Lurie, identifying the CY- A_3 obstruction group with $\text{HH}_{E_1}^{-2}(A, A)$.

Open problems.

- (i) *Non-abelian Dunn.* At non-simply-laced $\mathfrak{g}$, the Costello–Gaiotto–Yagi 5D hCS argument for the Yangian VOA all-orders theorem requires Dunn additivity for twisted affine algebras; the chain-level compatibility of Kontsevich–Tamarkin formality with the Dynkin-diagram twist is open.
- (ii) *Global-chart Dunn.* For a compact CY₃ X with non-trivial $\text{Aut}(X)$, does the local-to-global Dunn equivalence $\Phi_3^{\text{FA}}(X) \simeq \bigotimes_{\alpha} E_3\text{-glue}$ commute with Aut -action? For $K3 \times E$ the Aut is $\text{Aut}(K3) \times \mathbb{Z}^2$, and the compatibility is expected but not verified in full generality.
- (iii) *Dunn at genus.* The chain-level Dunn equivalence on the local model $\mathbb{R}^3$ lifts to a genus- g statement via Francis–Gaitsgory factorisation homology; the explicit $g = 1, 2, 3$ chain-level witnesses for the genus-tower identities of the braided factorization chapter (Propositions fh-genus-tower and the higher-genus $K3 \times E$ partition functions) are partially completed.
- (iv) *Dunn at roots of unity.* At $q = e^{2\pi i/N}$, the Dunn equivalence descends to a statement about N -braided operads (quantum-group / Lusztig-cell-cocycle variant); the chain-level comparison of root-of-unity Dunn with the tilting-mode quantum group at level N is open.

Next attack cycle.

- (Re-attack 1) Verify item (i): the explicit chain-level map $\mu: C_{\bullet}^{E_1} \otimes C_{\bullet}^{E_2} \rightarrow C_{\bullet}(\mathcal{C}_3; k)$ of Theorem 4.4 in a non-trivial chain-level model (Kontsevich graph complex).
- (Re-attack 2) Verify item (iii): compute the $g = 1$ Dunn witness for the $K3$ Heisenberg VOA, cross-check against the lattice θ -character.
- (Re-attack 3) Verify item (iv): check the Dunn descent at $q = e^{2\pi i/8}$ against the Lusztig reflection length $\ell_{\text{Lusztig}} = 8$ for the $K3$ Mukai lattice.

Five attack–heal cycles.

- Cycle 1:** *Attack.* The statement $E_3 \simeq E_1 \otimes E_2$ — does it hold in dg-operads over $\mathbb{Q}$, or only in the homotopy category? *Healing.* Holds in dg-operads up to quasi-isomorphism, as a consequence of Eilenberg–Zilber applied to the space-level equivalence (Theorem 4.1). The quasi-isomorphism class is canonical; the strict equality fails (the associator choice is a GRT_1 -torsor).
- Cycle 2:** *Attack.* The claim that Stage-2 specialisation consumes the E_2 -factor — is this specific to the Dunn decomposition chosen, or does every (Σ_2, C) consume a different E_2 -factor? *Healing.* Every (Σ_2, C) consumes the E_2 -factor aligned with its own transverse direction. Different choices consume different E_2 -factors, producing different E_1 -chiral outputs. The family of E_1 -chiral shadows parametrised by (Σ_2, C) is precisely the family of different consumptions.
- Cycle 3:** *Attack.* Is the Drinfeld-centre recovery of the transverse E_2 the same E_2 -structure that was consumed by Sp^{ch} ? *Healing.* No. The consumed E_2 -factor is the transverse Dunn factor on A itself; the recovered E_2 -structure is the Drinfeld-centre half-braiding on $\mathcal{Z}(\text{Rep}^{E_1}(\Phi_3(\mathcal{C})))$. These are related by Ben-Zvi–Francis–Nadler ($\mathcal{Z}(\text{Rep}^{E_1}(A)) \simeq \text{Rep}^{E_2}(\text{HH}^{\bullet}(A))$), but they are genuinely distinct E_2 -structures. See Theorem 8.1.

Cycle 4: *Attack.* The CFG 2026 physical picture uses a *topological* E_2 -factor (Kapustin–Witten twist); our chiral setting uses a *holomorphic* E_2 -factor. Can both be Dunn-factors of the same E_3 ? *Healing.* Yes, via the Wick-rotation equivalence of Theorem 9.1. At the operadic level $E_1 \otimes E_2$ does not distinguish topological from holomorphic; the distinction is in the interpretation of the operad action on observables. CFG §3.6 equates the two on shell.

Cycle 5: *Attack.* The CY- A_3 obstruction proof used the identification $L_{E_3/E_2}(A) = \Sigma^2 L_{E_1}(A)$ (Francis Theorem 4.1). Is this identification dependent on the choice of formality presentation? *Healing.* No. The relative cotangent complex is intrinsic to the $E_2 \rightarrow E_3$ forgetful functor, which is determined by Dunn–Lurie independent of formality. Different associators yield different strict cotangent models but the quasi-isomorphism class is the GRT_1 -invariant $\Sigma^2 L_{E_1}(A)$ (Theorem 10.1).